

**ECON 101—Introduction to Microeconomics
Spring 2026**

FINAL EXAM

April 30, 2026

General Instructions

- The exam has a duration of **110 minutes**. If you finish early, you may leave within the first 100 minutes only. Otherwise remain quietly until the exam concludes.
- All answers must be written in the answer sheet. Scratch work will not be considered. You may not take this booklet with you.
- **Allowed on your desk or person:** ID, writing instruments, ruler, and a simple calculator. **Nothing else.** No extra paper, no graphing calculators, no phones, no smart devices, no electronics, no unauthorized materials of any kind. Any item not explicitly listed above must be placed outside the room or at the front before the exam begins. No exceptions.
- You may request a bathroom break, but only one person may be out at a time.
- Crying is allowed, but quietly, and not on the exam paper.
- Good luck and may your choice of inputs maximize your profits!

Exam Instructions and Academic Dishonesty Policy

Before the exam begins:

- **Allowed on your desk or person:** ID, writing instruments, ruler, and a simple calculator. **If it is not on this list, it is not allowed.** All prohibited items must be placed outside the room or at the front before the exam begins.
- Wait until everyone is seated. Do **not** write anything—including your name and ID—until you are explicitly told to do so.

At the end of the exam:

- When time is called, put all writing instruments down immediately, close your booklet, keep your hands away from it, and remain seated until dismissed.

Academic Dishonesty:

- Any form of cheating, including any deviation from the instructions above, will be met with the **harshest consequences available**. There will be **no sympathy and no leniency**.

Answer Sheet Instructions

- **Record your exam version on the answer sheet.** If no version is indicated, the lowest-scoring version will be applied. No exceptions.
- **Use only the provided pencil and erase cleanly if you make a mistake.**
- This exam has 25 questions worth a total of 100 points plus one bonus question worth 3 points. Point values are indicated in parentheses and reflect perceived difficulty—by me, though you may think otherwise.

Exam Concept version

This exam has 25 questions, worth 3, 5 or 7 points depending on perceived difficulty, for a total of 100 points. You may disagree. There is also a bonus question worth 3 points.

Question permutation table:

Question	Version	1	2	3	4
1	→	11	39	33	21
2*	→	12	40	34	22
3	→	30	4	3	23
4*	→	31	5	4	24
5*	→	32	6	5	25
6	→	1	7	18	40
7	→	33	20	35	4
8*	→	34	21	36	5
9*	→	35	22	37	6
10*	→	36	23	38	7
11*	→	37	24	39	8
12*	→	38	25	40	9
13	→	2	33	2	10
14	→	24	31	25	20
15	→	25	8	15	19
16	→	13	19	13	27
17	→	14	32	6	11
18	→	9	3	16	2
19	→	40	34	20	1
20	→	3	35	7	30
21	→	10	14	17	17
22	→	26	1	8	3
23	→	4	15	9	31
24*	→	5	16	10	32
25*	→	6	17	11	33
26*	→	7	18	12	34
27	→	28	37	26	28
28*	→	29	38	27	29
29	→	15	10	21	12
30*	→	16	11	22	13
31*	→	17	12	23	14
32*	→	18	13	24	15
33	→	19	26	28	35
34*	→	20	27	29	36
35*	→	21	28	30	37
36*	→	22	29	31	38
37*	→	23	30	32	39
38	→	39	2	1	16
39	→	27	36	19	18
40	→	8	9	14	26

*: Question follows previous

Consider the production function:

$$F(K, L) = 3L^{1/2} + K^{1/2}$$

The following information pertains to questions 1 and 2.

1. What are the returns to scale?

- (a) Increasing returns to scale
- (b) Constant returns to scale
- (c) Decreasing returns to scale
- (d) Depends on K and L ratio

Correct answers: (c)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(c)	(a)	(b)	(c)
(b)	→	(b)	(b)	(c)	(d)
(c)	→	(a)	(c)	(a)	(a)
(d)	→	(d)	(d)	(d)	(b)

2. What happens to the Marginal Product of capital (MP_K) as K increases?

- (a) Increasing
- (b) Constant
- (c) Decreasing
- (d) First increasing, then decreasing

Correct answers: (c)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(a)	(b)	(c)	(c)
(b)	→	(b)	(c)	(d)	(b)
(c)	→	(c)	(a)	(a)	(a)
(d)	→	(d)	(d)	(b)	(d)

3. Anel is a baker at “Kink 101” bakery. She’s preparing a new glaze recipe and came up with the following:

$$g_A = \min\{s, 4c\},$$

where s are units of sugar, c are units of cream, and g_A are units of glaze according to Anel’s recipe. To produce 3 units of glaze, what is the minimum amount of cream Anel needs?

- (a) 3
- (b) $\frac{3}{4}$
- (c) $\frac{1}{4}$
- (d) 1

Correct answers: (b)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(a)	(d)	(b)	(d)
(b)	→	(c)	(b)	(a)	(c)
(c)	→	(d)	(c)	(d)	(b)
(d)	→	(b)	(a)	(c)	(a)

4. Malika also works at the bakery, but she thinks Anel’s glaze is too sweet, and that the perfect glaze recipe should be:

$$g_M = \min\{6s, c\},$$

where g_M are units of glaze according to Malika’s recipe. What is the minimum amount of sugar Malika needs to produce 3 units of glaze?

- (a) 3
- (b) $\frac{3}{4}$
- (c) 6
- (d) $\frac{1}{2}$

Correct answers: (d)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(d)	(a)	(b)	(a)
(b)	→	(c)	(b)	(d)	(c)
(c)	→	(a)	(c)	(c)	(d)
(d)	→	(b)	(d)	(a)	(b)

5. Medina, who is the owner of Kink 101 bakery, has to decide which glaze recipe will be used for all cakes. From her point of view, in the production of cakes, c , both recipes are perfect substitutes, e.g., $c = g_A + g_M$. Suppose that sugar costs 1 per unit and cream costs 10 per unit. Which recipe will Medina choose if she wants to minimize cost?

- (a) Anel's recipe
- (b) Malika's recipe
- (c) Medina is indifferent
- (d) Cannot be determined

Correct answers: (a)

Answer permutations: (All answers permuted, except last answer)

Version		1	2	3	4
Answer					
(a)	→	(c)	(a)	(c)	(b)
(b)	→	(b)	(b)	(a)	(c)
(c)	→	(a)	(c)	(b)	(a)
(d)	→	(d)	(d)	(d)	(d)

6. Akmaral is the owner of a garden with apples. She wants to sell apples this summer at the local market (assume the local market is competitive). Her friend Sultan helped her calculate the short-run cost, which is

$$C(q) = k + \frac{q^3}{2} + q.$$

What is the price of shutting down?

- (a) $p = 0$
- (b) $p = 1$
- (c) $p = k$
- (d) $p = \frac{3}{2}q^2 + 1$

Correct answers: (b)

Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(b)	(a)	(c)	(d)
(b)	→	(a)	(c)	(b)	(a)
(c)	→	(d)	(b)	(a)	(b)
(d)	→	(c)	(d)	(d)	(c)

Shugyla loves matcha latte from Drinkit. However, because she lives far away from the university, she cannot enjoy it every day. Nurai suggests preparing matcha latte at home using the following recipe:

- 2 teaspoons of matcha powder
- 1 cup of milk
- 2 teaspoons of hot water
- 1 teaspoon of sweetener

Let M = matcha powder, L = milk, W = hot water, S = sweetener. The following information pertains to questions 7–12.

7. According to the above recipe, what would be the appropriate production function to model the recipe that Nurai provided to Shugyla?

- (a) $q = M + L + W + S$
- (b) $q = \min(M, L, W, S)$
- (c) $q = \min\left(\frac{M}{2}, L, \frac{W}{2}, S\right)$
- (d) $q = \frac{M + L + W + S}{4}$

Correct answers: (c)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(c)	(d)	(d)	(c)
(b)	→	(b)	(a)	(c)	(d)
(c)	→	(a)	(b)	(b)	(b)
(d)	→	(d)	(c)	(a)	(a)

8. Suppose Shugyla owns only one frother machine, and with one frother she can make at most 6 lattes per hour regardless of ingredients. What is the new production function?

- (a) $q = \min\left(\frac{M}{2}, L, \frac{W}{2}, S, 6\right)$
 (b) $q = \max\left(\frac{M}{2}, L, \frac{W}{2}, S, 6\right)$
 (c) $q = \min\left(\frac{M}{2}, L, \frac{W}{2}, S\right) + 6$
 (d) $q = 6 \min(M, L, W, S)$

Correct answers: (a)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(c)	(d)	(a)	(c)
(b)	→	(d)	(a)	(c)	(b)
(c)	→	(b)	(b)	(b)	(d)
(d)	→	(a)	(c)	(d)	(a)

9. If Shugyla has enough ingredients for 5 lattes, but only one frother, how many can she actually produce in one hour?

- (a) 3
 (b) 5
 (c) 6
 (d) 11

Correct answers: (b)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(c)	(c)	(b)	(b)
(b)	→	(a)	(d)	(c)	(a)
(c)	→	(d)	(b)	(a)	(c)
(d)	→	(b)	(a)	(d)	(d)

10. Ignoring the frother constraint from questions 8 and 9, suppose sweetener is no longer required. What is the new production function?

- (a) $q = \min\left(\frac{M}{2}, L, \frac{W}{2}\right)$
 (b) $q = \min\left(\frac{M}{2}, L, \frac{W}{2}, S\right)$
 (c) $q = M + L + W$
 (d) $q = \frac{M}{2} + L + \frac{W}{2}$

Correct answers: (a)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(d)	(a)	(b)	(b)
(b)	→	(b)	(c)	(d)	(d)
(c)	→	(a)	(d)	(a)	(c)
(d)	→	(c)	(b)	(c)	(a)

11. Now suppose Shugyla considers only matcha powder and milk as inputs. She wants to produce 5 matcha lattes and can choose among three technologies:

- Technology 1: 5 tsp matcha powder + 5 cups milk
- Technology 2: 5 tsp matcha powder + 3 cups milk
- Technology 3: 5 tsp matcha powder + 2 cups milk

If she can freely choose which technological process to use, which verbal description below represents the correct description of the combined isoquant for producing 5 matcha lattes?

- (a) A single L-shaped isoquant at (5, 5) representing Technology 1 only.
- (b) A smooth convex curve connecting the points (5, 5), (5, 3), and (5, 2).
- (c) Three disconnected L-shaped isoquants for each technology, without any connection.
- (d) A single L-shaped isoquant at (5, 2) representing Technology 3 only.

Correct answers: (d)

Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(d)	(d)	(a)	(a)
(b)	→	(a)	(b)	(b)	(c)
(c)	→	(c)	(a)	(c)	(d)
(d)	→	(b)	(c)	(d)	(b)

12. Suppose milk costs 400 tenge per cup, while matcha powder costs 100 tenge per teaspoon. Which technology minimizes total cost for producing 5 lattes?

- (a) Technology 1
- (b) Technology 2
- (c) Technology 3
- (d) All technologies have the same cost

Correct answers: (c)

Answer permutations: (All answers permuted, except last answer)

Version		1	2	3	4
Answer					
(a)	→	(b)	(b)	(b)	(b)
(b)	→	(a)	(c)	(a)	(c)
(c)	→	(c)	(a)	(c)	(a)
(d)	→	(d)	(d)	(d)	(d)

13. In the short run, which of the following is true?

- (a) All inputs are variable.
- (b) At least one input is fixed.
- (c) All costs are sunk.
- (d) The firm can freely enter or exit the market.

Correct answers: (b)

Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(a)	(b)	(c)	(c)
(b)	→	(d)	(d)	(b)	(d)
(c)	→	(c)	(c)	(d)	(a)
(d)	→	(b)	(a)	(a)	(b)

14. The law of diminishing marginal productivity states that as a firm increases one input while holding all others constant, the marginal product of that input will eventually:
- (a) Increase at an increasing rate
 - (b) Remain constant
 - (c) Decrease
 - (d) Become negative immediately

Correct answers: (c)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(d)	(d)	(c)	(a)
(b)	→	(a)	(c)	(b)	(b)
(c)	→	(b)	(a)	(a)	(d)
(d)	→	(c)	(b)	(d)	(c)

15. The marginal rate of technical substitution (MRTS) of capital for labor is equal to:

- (a) $\frac{MP_K}{MP_L}$
- (b) $\frac{MP_L}{MP_K}$
- (c) $\frac{AP_L}{AP_K}$
- (d) $\frac{K}{L}$

Correct answers: (b)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(c)	(b)	(d)	(b)
(b)	→	(d)	(a)	(c)	(a)
(c)	→	(a)	(c)	(b)	(d)
(d)	→	(b)	(d)	(a)	(c)

16. A programmer quits a job paying \$80,000 per year to start her own firm. In the first year, her firm earns \$100,000 in revenue and incurs \$30,000 in explicit costs. Her economic profit is:
- (a) \$70,000
 - (b) \$100,000
 - (c) \$20,000
 - (d) -\$10,000

Correct answers: (d)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(a)	(d)	(d)	(d)
(b)	→	(d)	(b)	(c)	(b)
(c)	→	(c)	(a)	(a)	(c)
(d)	→	(b)	(c)	(b)	(a)

17. Which of the following statements is correct?

- (a) In the long run, all costs are sunk.
- (b) Sunk costs vary with the level of output.
- (c) In the long run, there are no sunk costs, but there may be fixed costs.
- (d) Variable costs cannot be avoided by shutting down.

Correct answers: (c)

Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(c)	(c)	(c)	(a)
(b)	→	(a)	(d)	(d)	(c)
(c)	→	(b)	(a)	(b)	(b)
(d)	→	(d)	(b)	(a)	(d)

18. At the cost-minimizing input combination in the long run, the firm sets:

- (a) $MP_L = MP_K$
- (b) $w = r$
- (c) $MRTS = \frac{w}{r}$
- (d) $\frac{w}{MP_K} = \frac{r}{MP_L}$

Correct answers: (c)

Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(c)	(c)	(c)	(a)
(b)	→	(d)	(a)	(a)	(b)
(c)	→	(a)	(d)	(b)	(d)
(d)	→	(b)	(b)	(d)	(c)

19. Short-run total costs are, in general, greater than or equal to long-run total costs for a given level of output because:

- (a) Short-run production functions are different from long-run production functions.
- (b) In the short run, at least one input is fixed, preventing the firm from choosing the cost-minimizing input combination.
- (c) Sunk costs are higher in the short run.
- (d) Input prices are higher in the short run.

Correct answers: (b)

Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(d)	(c)	(d)	(b)
(b)	→	(a)	(b)	(a)	(d)
(c)	→	(b)	(a)	(b)	(c)
(d)	→	(c)	(d)	(c)	(a)

20. A profit-maximizing competitive firm should increase output whenever:

- (a) Price exceeds average total cost.
- (b) Marginal revenue exceeds marginal cost.
- (c) Total revenue exceeds total cost.
- (d) Average revenue exceeds average variable cost.

Correct answers: (b)

Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(b)	(b)	(d)	(c)
(b)	→	(c)	(c)	(c)	(a)
(c)	→	(d)	(d)	(a)	(b)
(d)	→	(a)	(a)	(b)	(d)

21. In the short run, a competitive firm should shut down if and only if:

- (a) It is making negative economic profits.
- (b) Price is below average total cost.
- (c) Revenue is less than avoidable costs.
- (d) Price is below marginal cost.

Correct answers: (c)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(b)	(c)	(d)	(c)
(b)	→	(c)	(a)	(b)	(b)
(c)	→	(d)	(d)	(a)	(a)
(d)	→	(a)	(b)	(c)	(d)

22. In a constant-cost industry with free entry and exit, the long-run competitive equilibrium is characterized by:

- (a) Firms earning positive economic profits.
- (b) Price equal to the minimum of long-run average total cost.
- (c) Each firm producing where marginal cost is minimized.
- (d) A downward-sloping long-run supply curve.

Correct answers: (b)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(b)	(d)	(c)	(a)
(b)	→	(d)	(b)	(a)	(c)
(c)	→	(c)	(c)	(b)	(b)
(d)	→	(a)	(a)	(d)	(d)

Use the following information to answer questions 23–26. A profit-maximizing, price-taking firm produces output, widgets, y using a single input, labor, x . The wage rate of x is \$450 per unit. The firm can produce 0, 8, 9, or 16 units of y using 0, 4, 7, or 9 units of x , respectively. There are no other input-output combinations. The firm production function is therefore given by:

$$y = f(x) = \begin{cases} 0 & \text{if } x = 0, \\ 8 & \text{if } x = 4, \\ 9 & \text{if } x = 7, \\ 16 & \text{if } x = 9. \end{cases}$$

23. What is the cost function $C(y)$ of this firm?

$$(a) \ C(y) = \begin{cases} 0 & \text{if } y = 0, \\ 1,800 & \text{if } y = 8, \\ 3,150 & \text{if } y = 9, \\ 4,050 & \text{if } y = 16. \end{cases}$$

$$(b) \ C(y) = \begin{cases} 0 & \text{if } y = 0, \\ 1,800 & \text{if } y = 8, \\ 3,150 & \text{if } y = 9, \\ 4,500 & \text{if } y = 16. \end{cases}$$

$$(c) \ C(y) = \begin{cases} 0 & \text{if } y = 0, \\ 2,000 & \text{if } y = 8, \\ 3,500 & \text{if } y = 9, \\ 4,050 & \text{if } y = 16. \end{cases}$$

$$(d) \ C(y) = \begin{cases} 0 & \text{if } y = 0, \\ 2,200 & \text{if } y = 8, \\ 3,150 & \text{if } y = 9, \\ 4,050 & \text{if } y = 16. \end{cases}$$

Correct answers: (a)

Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(c)	(b)	(b)	(d)
(b)	→	(a)	(d)	(d)	(b)
(c)	→	(b)	(a)	(a)	(a)
(d)	→	(d)	(c)	(c)	(c)

24. Let p be the price of a widget. What is the firm's supply function $y(p)$?

$$(a) \ y(p) = \begin{cases} 0 & \text{if } 0 \leq p < 225, \\ 8 & \text{if } 225 \leq p < 253.125, \\ 9 & \text{if } 253.125 \leq p < 315, \\ 16 & \text{if } p \geq 315. \end{cases}$$

$$(b) \ y(p) = \begin{cases} 0 & \text{if } 0 \leq p < 200, \\ 8 & \text{if } 200 \leq p < 253.125, \\ 16 & \text{if } p \geq 253.125. \end{cases}$$

$$(c) \ y(p) = \begin{cases} 0 & \text{if } 0 \leq p < 225, \\ 8 & \text{if } 225 \leq p < 253.125, \\ 16 & \text{if } p \geq 253.125. \end{cases}$$

$$(d) \ y(p) = \begin{cases} 0 & \text{if } 0 \leq p < 225, \\ 8 & \text{if } 225 \leq p < 253.125, \\ 9 & \text{if } 253.125 \leq p < 300, \\ 16 & \text{if } p \geq 300. \end{cases}$$

Correct answers: (c)

Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(c)	(b)	(d)	(c)
(b)	→	(a)	(a)	(c)	(d)
(c)	→	(d)	(c)	(a)	(a)
(d)	→	(b)	(d)	(b)	(b)

25. Suppose that there are 5 such price taking firms. What is the industry supply $Y^S(p)$?

$$\begin{aligned}
 \text{(a) } Y^S(p) &= \begin{cases} 0 & \text{if } 0 \leq p < 225, \\ 40 & \text{if } 225 \leq p < 253.125, \\ 45 & \text{if } 253.125 \leq p < 315, \\ 80 & \text{if } p \geq 315. \end{cases} \\
 \text{(b) } Y^S(p) &= \begin{cases} 0 & \text{if } 0 \leq p < 200, \\ 40 & \text{if } 200 \leq p < 253.125, \\ 80 & \text{if } p \geq 253.125. \end{cases} \\
 \text{(c) } Y^S(p) &= \begin{cases} 0 & \text{if } 0 \leq p < 225, \\ 40 & \text{if } 225 \leq p < 253.125, \\ 80 & \text{if } p \geq 253.125. \end{cases} \\
 \text{(d) } Y^S(p) &= \begin{cases} 0 & \text{if } 0 \leq p < 225, \\ 40 & \text{if } 225 \leq p < 253.125, \\ 45 & \text{if } 253.125 \leq p < 300, \\ 80 & \text{if } p \geq 300. \end{cases}
 \end{aligned}$$

Correct answers: (c)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(c)	(c)	(c)	(b)
(b)	→	(a)	(b)	(d)	(a)
(c)	→	(b)	(d)	(a)	(d)
(d)	→	(d)	(a)	(b)	(c)

26. Suppose the market demand function is $Y^D(p) = 100 - \frac{p}{45}$. What is the market equilibrium?

- (a) $p^* = 225$; $Y^* = 95$ units
- (b) $p^* = 253.125$; $Y^* = 94.375$ units
- (c) $p^* = 900$; $Y^* = 80$ units
- (d) $p^* = 450$; $Y^* = 90$ units

Correct answers: (c)

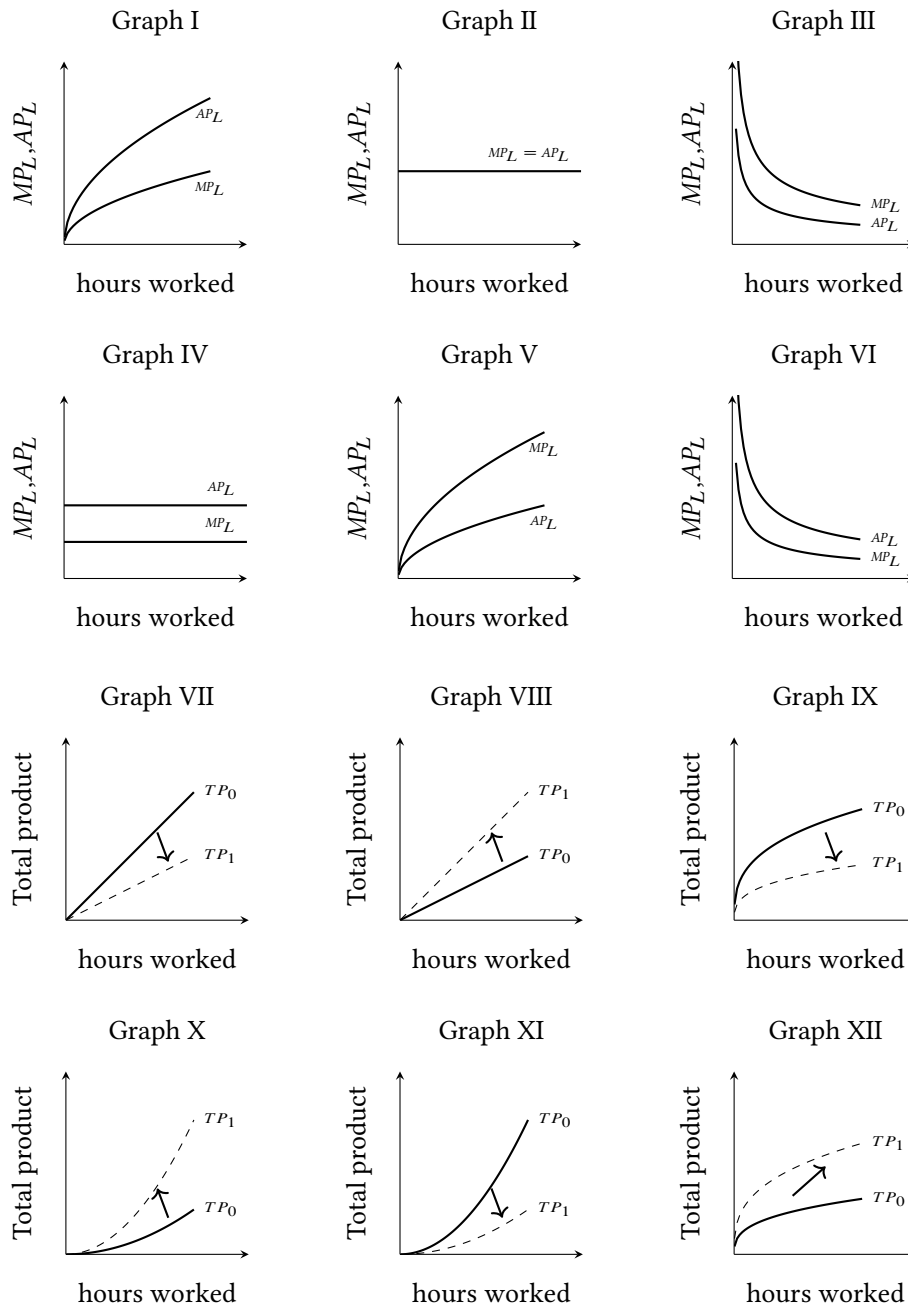
Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(b)	(b)	(a)	(a)
(b)	→	(d)	(a)	(d)	(d)
(c)	→	(c)	(c)	(c)	(c)
(d)	→	(a)	(d)	(b)	(b)

Analyze the graphs below and answer questions 27 and 28.

Entrepreneur A claims that each additional worker they hire produces the same number of units as the previous worker. Entrepreneur B claims that each additional worker they hire produces fewer units than the previous worker. Entrepreneur C claims that each additional worker they hire produces more units than the immediately

preceding worker.



27. The graph _____ represents the average and marginal product of Entrepreneur A. The graph _____ represents the average and marginal product of Entrepreneur B. The graph _____ represents the average and marginal product of Entrepreneur C.

- (a) II; VI; V.
- (b) IV; VI; V.
- (c) II; III; I.
- (d) I; II; III.

Correct answers: (a)

Answer permutations: (All answers permuted)

Version	1	2	3	4
Answer				
(a)	→ (d)	(a)	(b)	(d)
(b)	→ (a)	(c)	(d)	(a)
(c)	→ (b)	(b)	(a)	(b)
(d)	→ (c)	(d)	(c)	(c)

28. If the three companies invest in a course on worker health care and on-the-job training, the graph _____ represents Company A, the graph _____ represents Company B, and the graph _____ represents Company C.

- (a) VII; IX; XI
- (b) VIII; XII; X
- (c) VI; VII; XII
- (d) VIII; X; XII

Correct answers: (b)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(d)	(d)	(c)	(a)
(b)	→	(a)	(c)	(a)	(c)
(c)	→	(c)	(b)	(b)	(b)
(d)	→	(b)	(a)	(d)	(d)

Consider an industry that has free entry and exit. Suppose that initially it is in a long-run equilibrium with constant costs with a fixed number of firms, and zero profits. Furthermore, suppose that firms are identical, and that they have upward sloping marginal costs. With this information, answer questions 29–32.

29. In the short-run, with a fixed number of firms, the supply curve is:

- (a) Upward sloping
- (b) Horizontal and perfectly elastic
- (c) Vertical and perfectly inelastic
- (d) None of the above.

Correct answers: (a)

Answer permutations: (All answers permuted, except last answer)

	Version	1	2	3	4
Answer					
(a)	→	(c)	(c)	(a)	(a)
(b)	→	(b)	(a)	(b)	(c)
(c)	→	(a)	(b)	(c)	(b)
(d)	→	(d)	(d)	(d)	(d)

30. In the long-run, the long-run aggregate supply curve is:

- (a) Horizontal and equal to the minimum average cost.
- (b) Upward sloping and equal to the minimum average cost which is increasing.
- (c) Downward sloping and equal to the minimum average cost which is decreasing.
- (d) I don't know and perhaps that is why I tried to pay someone to take the exam for me.

Correct answers: (a)

Answer permutations: (All answers permuted, except last answer)

	Version	1	2	3	4
Answer					
(a)	→	(b)	(b)	(b)	(c)
(b)	→	(c)	(a)	(c)	(a)
(c)	→	(a)	(c)	(a)	(b)
(d)	→	(d)	(d)	(d)	(d)

31. Now suppose that there is an increase in demand for the good that this industry produces. Then after the adjustment to the new equilibrium in the short-run:
- (a) The new equilibrium price will be higher compared to the original equilibrium price; producers will be making economic profits.
 - (b) The new equilibrium price will be lower compared to the original equilibrium price; producers will be making economic profits.
 - (c) The new equilibrium price will be higher compared to the original equilibrium price; producers will be making economic losses.
 - (d) The new equilibrium price will be lower compared to the original equilibrium price; producers will be making economic losses.

Correct answers: (a)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(a)	(b)	(a)	(a)
(b)	→	(b)	(d)	(c)	(d)
(c)	→	(c)	(a)	(b)	(c)
(d)	→	(d)	(c)	(d)	(b)

32. After the increase in the demand and the short-run adjustment, suppose that the government restricts entry into the industry by imposing a strict control on entry via operation permits. Then, in the long-run, the long-run aggregate supply is:
- (a) Horizontal and equal to the minimum average cost.
 - (b) Upward sloping due to the imposition of the operation permits.
 - (c) Downward sloping due to the imposition of the operation permits.
 - (d) I don't know because instead of studying I was trying to find someone who could take the exam for me.

Correct answers: (b)

Answer permutations: (All answers permuted, except last answer)

	Version	1	2	3	4
Answer					
(a)	→	(a)	(a)	(b)	(c)
(b)	→	(b)	(c)	(a)	(a)
(c)	→	(c)	(b)	(c)	(b)
(d)	→	(d)	(d)	(d)	(d)

The following information pertains to questions 33–37. Prof. A (for anonymous) and one of his past TAs, Sabina, are thinking to open a cat cafe in Astana, where customers can buy food for their cats, coffee for themselves, or just play and have fun with the cats that live there. Since Sabina passed her BUS101 course with flying colors, Prof. A decides to partner with her in this new business venture. However, Sabina, having taken ECON101 before, understands the very important need to make a detailed economic analysis. It turns out that the production function of a cat cafe can be modeled as follows: Let h be the total number of happy customers served in the cafe, then we have that

$$h = F(L, K, Z) = L^{2/3} K^{1/3} Z,$$

where L is hours of labor from the cat cafe attendant, K is the number of cats in the cafe, and Z is a variable that can either be 1 or 0, and denotes whether the fixed costs of the cafe are paid (rent/lease, renovation, permits, etc.); notice that production cannot happen unless these fixed costs are paid. Sabina estimates that rent per month, plus renovation, furnishing, and monthly repair costs, plus licensing and permits would amount to a total of ₸700,000 per month, i.e., $p_Z = 700,000$. In order to determine the other costs, Sabina and Prof. A decide to hire Daniyar as a consultant, who is an authority in *creative accounting* as well as a certified cat expert. Daniyar estimates that the monthly cost of maintaining a single cat (dry food, wet food, veterinary care, grooming, insurance, basic needs, and toys) would be around ₸100,000 per cat, per month, i.e., $p_K = 100,000$.¹ Sabina and Prof. A also decide to hire Daniyar as the attendant of the cat cafe, since he loves cats, is very good at handling difficult clients, and his

¹You may think this is very expensive, but Prof. A instructed Daniyar to estimate this number based on cats that are treated as royalty, as it should be.

career as a banker didn't work out. They decide to pay Daniyar a wage rate per hour of ₹5,000, i.e., $w = 5,000$.

33. Suppose that Prof. A and Sabina decide to go forward, and *sink the investment of* paying the cost of Z , and start the cat cafe with $K = 8$ cats. If they want to serve h customers, what is the short-run total cost function of the cafe?

- (a) $SRTC(h) = \frac{2500}{2\sqrt{2}}h^{3/2} + 1,500,000$
 (b) $SRTC(h) = \frac{2500}{\sqrt{2}}h^{3/2} + 1,500,000$
 (c) $SRTC(h) = \frac{2500}{\sqrt{2}}h^{3/2} + 700,000$
 (d) $SRTC(h) = \frac{2500}{\sqrt{2}}h^{3/2} + 700,000$

Correct answers: (b)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(b)	(b)	(d)	(d)
(b)	→	(c)	(c)	(b)	(b)
(c)	→	(d)	(d)	(a)	(a)
(d)	→	(a)	(a)	(c)	(c)

34. Suppose that p_h is total expenditure per customer. From the point of view of the short run, what is the minimal level of p_h that each customer should spend so that Prof. A and Sabina can afford to keep Daniyar on the payroll? What is the minimal level of p_h that each customer should spend so that they break even? *Hint: you may find this information useful: $\frac{1}{4}x^{3/2} = (3/2)\sqrt{x}$, although this is not the only way to solve this problem.*

- (a) For the shutdown price, 0; for the break-even price, 31,628.7
 (b) 0 for both.
 (c) 0 and 24,533
 (d) I don't know, maybe I should also have paid attention to ECON101 like Sabina.

Correct answers: (a)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(a)	(b)	(c)	(d)
(b)	→	(c)	(d)	(d)	(a)
(c)	→	(b)	(c)	(a)	(b)
(d)	→	(d)	(a)	(b)	(c)

35. Being an excellent ECON101 student, Sabina recognizes that the decision to open the cafe should also be made before deciding to sink any investment, i.e., they need to solve the long-run problem. Suppose that due to capacity constraints, they cannot serve more than 1000 customers per month. If they want to serve h customers, what is the long-run total cost function of the cafe? You will need the following: if $Z = 1$, $MP_L = \frac{2}{3}\left(\frac{K}{L}\right)^{1/3}$, and $MP_K = \frac{1}{3}\left(\frac{L}{K}\right)^{2/3}$.

- (a) $LRTC(h) = 15,000(5)^{1/3}h + 700,000$.
 (b) $LRTC(h) = 15,000(5)^{1/3}h + 1,500,000$ for $h > 0$, $LRTC(0) = 0$.
 (c) $LRTC(h) = 15,000(5)^{1/3}h + 700,000$ for $h > 0$, $LRTC(0) = 0$, vertical at $h = 1,000$.
 (d) $LRTC(h) = 15,000(5)^{1/3}h + 700,000$ for $h > 0$, $LRTC(0) = 0$.

Correct answers: (d)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(c)	(a)	(d)	(c)
(b)	→	(a)	(b)	(c)	(d)
(c)	→	(b)	(d)	(a)	(b)
(d)	→	(d)	(c)	(b)	(a)

36. Once again, suppose that p_h is the total expenditure per customer. From the point of view of the long run, what is the minimal level of p_h that each customer should spend so that Prof. A and Sabina break even?

- (a) 0
- (b) 24,533
- (c) 26,349.6
- (d) 10,000

Correct answers: (c)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(b)	(d)	(d)	(a)
(b)	→	(c)	(c)	(b)	(c)
(c)	→	(d)	(b)	(a)	(b)
(d)	→	(a)	(a)	(c)	(d)

37. Suppose that there are already a few cat cafes in Astana, that the market is competitive, and that at the current market equilibrium, the average expenditure per customer in each cafe is ₸28,000. Suppose that if they enter the market, the equilibrium price will fall to ₸27,000. Should Prof. A and Sabina open the cat cafe?

- (a) No :(
- (b) Yes :)
- (c) I don't know, so I definitively should have followed Sabina's steps and study more for my ECON101 final.

Correct answers: (b)

Answer permutations: (All answers permuted, except last answer)

	Version	1	2	3	4
Answer					
(a)	→	(a)	(a)	(b)	(a)
(b)	→	(b)	(b)	(a)	(b)
(c)	→	(c)	(c)	(c)	(c)

Consider a firm with the following production function:

$$Q = A \cdot L^{0.3} K^{0.3} M^{0.2}$$

where:

- Q is the quantity of output,
- A is total factor productivity,
- L is the quantity of labor input,
- K is the quantity of capital input,
- M is the quantity of material input.

38. Suppose the firm increases all inputs by a constant factor $t > 1$. Which of the following statements correctly describes the returns to scale of this production function?

- (a) The production function exhibits **increasing returns to scale**.
- (b) The production function exhibits **constant returns to scale**.
- (c) The production function exhibits **decreasing returns to scale**.
- (d) The production function exhibits **variable returns to scale** depending on the value of t .

Correct answers: (c)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(b)	(a)	(a)	(b)
(b)	→	(c)	(c)	(d)	(c)
(c)	→	(d)	(b)	(b)	(a)
(d)	→	(a)	(d)	(c)	(d)

Consider the following two statements:

Statement 1: Consider the market for cars. All firms have the same production technology, and we know that the long-run average cost curve has a minimum at a cost of 4,600,000 Tenge per car. Therefore, we can never observe a market equilibrium in which the equilibrium price is below 4,600,000 Tenge per car.

Statement 2: Consider a firm that produces computers. Suppose that the market price of a computer is $\bar{p}$, both in the short run and the long run. If in the short run this firm decides to shut down, then it must be that in the long run it decides to exit the market.

39. Which of the following options correctly describes the truth values of the statements?

- (a) Both statements are **True**.
- (b) Statement 1 is **True**; Statement 2 is **False**.
- (c) Statement 1 is **False**; Statement 2 is **True**.
- (d) Both statements are **False**.

Correct answers: (d)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(c)	(c)	(d)	(a)
(b)	→	(b)	(b)	(c)	(b)
(c)	→	(a)	(a)	(a)	(d)
(d)	→	(d)	(d)	(b)	(c)

40. (Bonus) What is the name of the teaching assistants leading your tutorials?

- (a) Qyzbolsyn, Wifibek, Chalkabay, Qoltyqshash, Gigachad, Sahur, Uwuzhan, Qyzbolar, Paluan
- (b) Assel, Shugyla, Tantemir, Manas, Nurailym, Damir, Alima, Arailym, Merey
- (c) Aruzhan, Tamerlan, Malika, Shugyla, Anel, Mansur, Nurai, Daryn, Almira
- (d) Anel, Shynar, Daniyal, Aruzhan, Nurailym, Malika, Manat, Temirkhan, Aidana, Altynay
- (e) I was too cool to attend tutorials

Correct answers: (c)

Answer permutations: (All answers permuted, except last answer)

	Version	1	2	3	4
Answer					
(a)	→	(a)	(a)	(d)	(d)
(b)	→	(d)	(d)	(b)	(a)
(c)	→	(c)	(b)	(c)	(b)
(d)	→	(b)	(c)	(a)	(c)
(e)	→	(e)	(e)	(e)	(e)